

■ Cludtica SaaS end-to-end Data Analysis



PYTHON, SQL, POWER BI

Created by : Snehal Prasad

BUSINESS PROBLEM



- Cloudtica a B2B SaaS company, lacked a centralized analytics system to monitor sales performance, profitability, customer retention, and regional growth. The company needed a data-driven solution to identify high-performing products, optimize discount strategies, improve customer retention, and support strategic business decisions through interactive dashboards and business insights.

DATASET OVERVIEW



Raw Dataset

Raw Dataset



Cleaned Dataset

Cleaned Dataset



Data Extract

- sourced data from Kaggle

Data Transformation

Clean and preprocess data using Python (Pandas and NumPy).
Handle missing values, corrected data types, standardized formats and removed inconsistencies.

Exploratory Data Analysis (EDA)

- Understand key patterns in sales, profits, discounts, and customer segments.
- Performed Descriptive Statistics Outlier, Detection (IQR Method) and Correlation Analysis

Data Warehousing

- Loaded Cleaned data into MySQL database
- Executed analytical SQL queries to extract insights from the dataset

Business Intelligence Dashboard

- Designed and Developed an interactive Power BI Dashboard to monitor key business metrics.
- Delivered actionable visual insights to support data-driven decision-making.

Tech Stack

Python

Data Cleaning and EDA

Tools & Libraries: Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook

- Data loading and preprocessing
- Handling missing values
- Removing duplicates
- Outlier detection and treatment
- Data transformation and feature creation

SQL

Data Storage & Business Queries

Database used: MySQL

- Import cleaned dataset into SQL database
- Perform multiple analytical queries
- Simulate real-world business data analysis scenarios

Power BI

Data Visualization & Dashboard

DAX (Data Analysis Expression) – Created multiple DAX measures used for calculated memory, dynamic visuals, and conditional logic.

- KPI cards
- Dynamic slicers
- Cross filtering
- Trend analysis
- Interactive visuals

Dashboard Screenshot

KPIs

Slicers

Positive upward sales trend indicating growing customer demand and successful sales initiatives

Products such as Site Analytics, FinanceHub, Marketing Suite – Gold generate the highest sales. Whereas, several products contribute relatively low revenue, indicating product optimization, targeted marketing.

The AMER region generated the highest sales(~\$192K), significantly outperforming EMEA and APJ. This suggests that AMER is the company's strongest market and should remain a key focus area.



Dashboard Screenshot



The "Top 10 Product by Profit" chart shows that Support generates the highest profit, followed by ContactMatcher, making them key revenue-driving products.

This chart indicates that Support leads in sales volume, suggesting strong customer demand and market adoption.



Summary

Product Analysis

Customer Analysis

Year

2020

2021

2022

2023

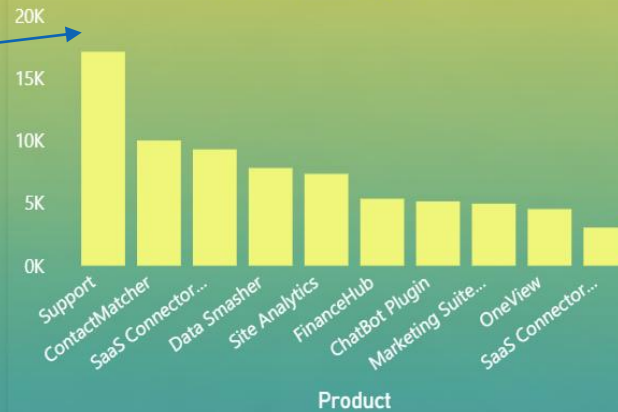
Segment

Enterprise

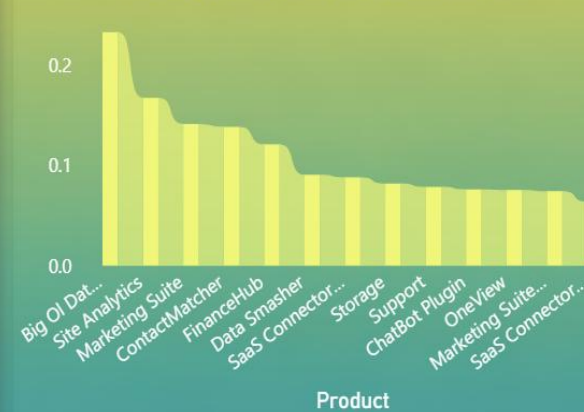
SMB

Strategic

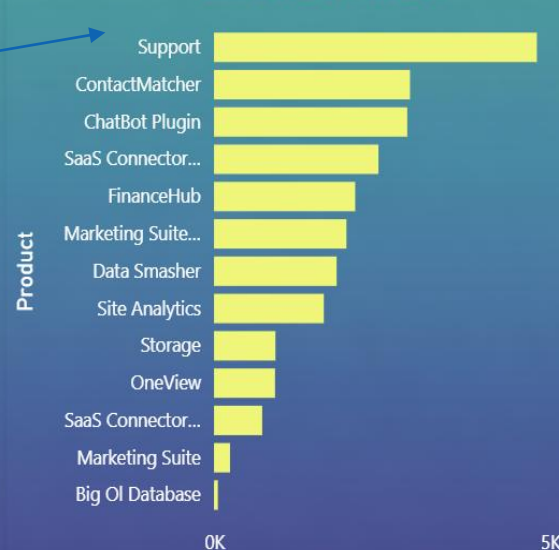
Top 10 Product by Profit



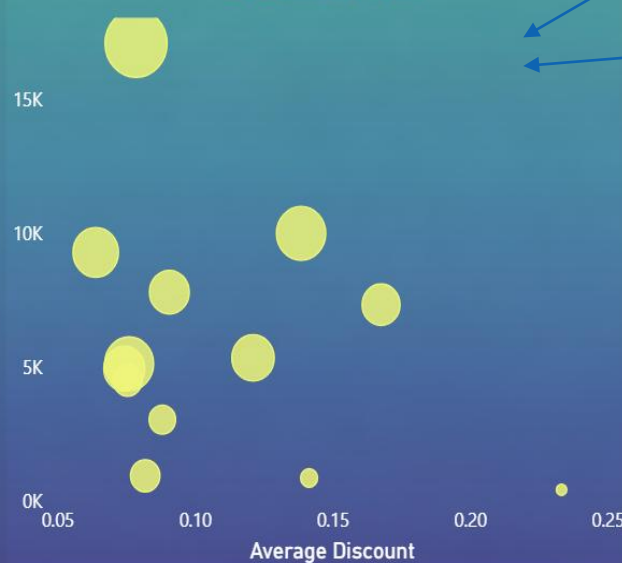
Average Discount Across Products



Units Sold by Product



Discount vs Profit



Region

AMER

APJ

EMEA

Industry

Communications

Consumer Produ...

Energy

Finance

Product

Big Ol Database

ChatBot Plugin

ContactMatcher

Data Smasher

Filters

Bubble Chart

Higher discounts do not always increase profit
From this "Discount vs Profit" bubble chart, products with moderate discounts achieve better profits, while very high discounts are associated with lower profitability.

Dashboard Screenshot



Summary

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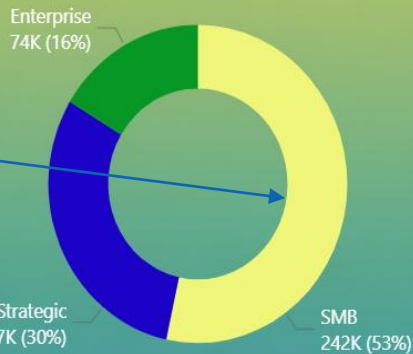
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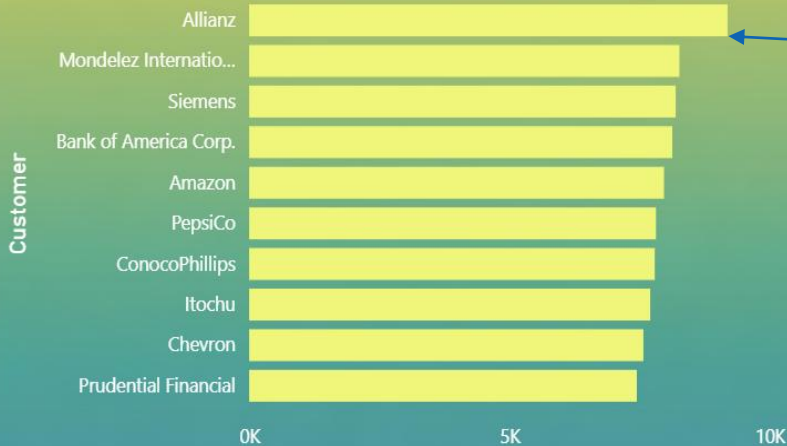
SMB

Strategic

Sales by Segment



Top 10 Customers by Sales



Region

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EMEA

Industry

Communications

Consumer Produ...

Energy

Finance

Product

Big Ol Data

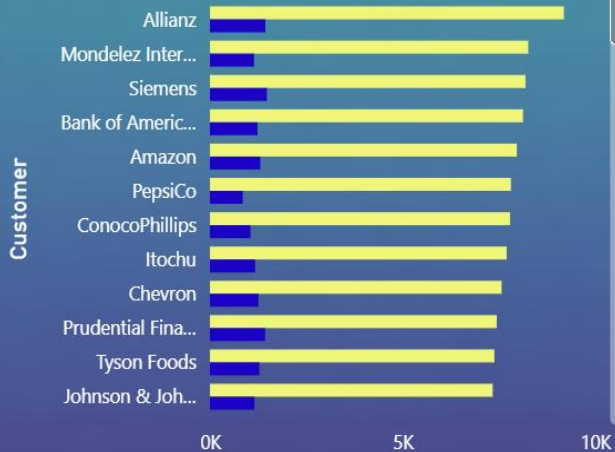
ChatBot

Contact

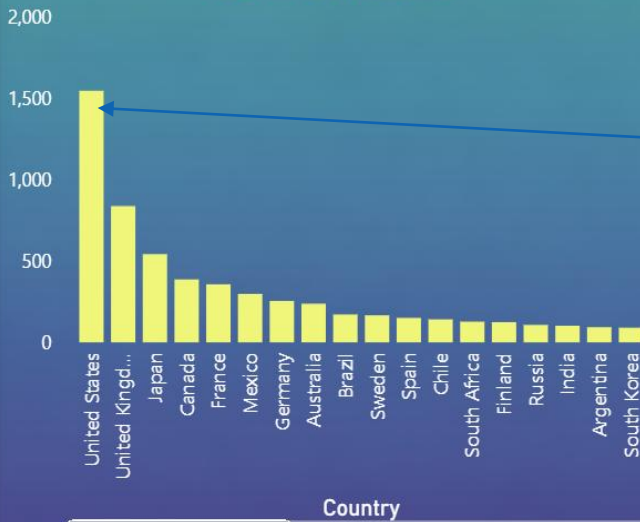
Data Smasher

Customer Profitability

● Total Sales ● Total Profit



Customers by Country



Allianz is the highest revenue-generating customer, followed by Mondelez International and Siemens. They contribute a significant share of total sales. Losing any of these customers could have a impact on overall revenue.

The Unites States has the highest number of customers, significantly exceeding all other countries. This suggest strong market penetration in the U.S. and potential growth opportunities in the other countries to diversify revenue sources.

Business Impact & Key Insights



Product Strategy

This project identifies which SaaS products generate the highest revenue and profit, helping the company focus on high-performing offerings.

Pricing & Discount Optimization

By analysing the relationship between discounts and profit, the company can refine discount strategies to avoid unnecessary margin loss.

Customer Relationship Management

Identifying top customers allows the company to strengthen relationships with high-value clients and improve retention strategies.

Regional Expansion

Understanding regional sales distribution helps determine where the company should expand its marketing and sales efforts.

Market Segmentation

Insights into industry and segment performance allow the company to better target high-potential markets.

SMB Segment Dominates Sales

The SMB segment contributes the largest share of total sales, indicating strong demand from small and medium-sized businesses.

Certain Products Generate Higher Profit Margins

Products such as Support and ContactMatcher show higher profitability compared to others, indicating strong pricing power.

Discounts Impact Profitability

Higher discounts do not always result in higher sales and often reduce profit margins, highlighting the need for optimized discount strategies.

Regional Sales Distribution

The AMER region contributes the largest portion of sales, followed by EMEA and APJ, indicating strong market presence in North America.

Revenue Concentration in Top Customers

- A small number of customers generate a significant share of total revenue, emphasizing the importance of maintaining strong relationships with key accounts.

Recommendations



Optimize Discount Strategy:

The company should carefully manage discount levels to maintain profitability while still attracting customers.

Focus on High-Profit Products:

Products generating the highest profit margins should receive increased marketing and sales focus.

Strengthen Relationships with Key Customers:

Since top customers contribute significantly to revenue, implementing loyalty programs and dedicated account management could improve retention.

Expand in High-Growth Regions:

Regions showing consistent growth should be prioritized for expansion and targeted marketing campaigns.

Industry Targeting:

Industries generating strong revenue should receive specialized product bundles or tailored SaaS solutions.

CONCLUSION



- This project demonstrates how raw business data can be transformed into actionable insights using a complete data analytics workflow. Through data cleaning, exploratory data analysis, database querying and interactive visualization, this project provides a comprehensive view of sales performance, product profitability, and customer behaviour. The Power BI dashboard enables stakeholders to monitor key performance indicators, identify growth opportunities, and make data-driven business decisions. Overall, this project highlights the importance of data-driven decision making in SaaS B2B businesses and showcases practical skills in Python, SQL, and Power BI.

THANK YOU

